SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

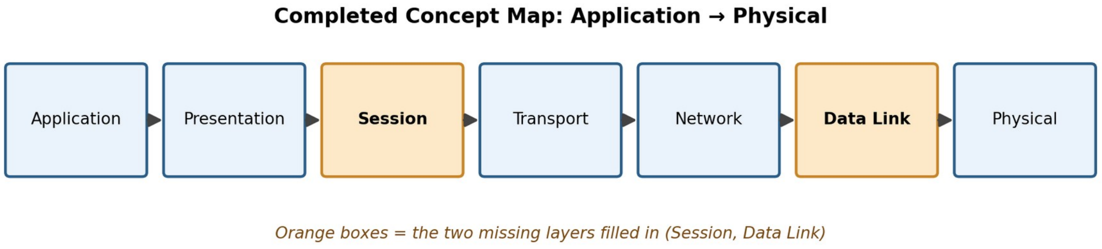
I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

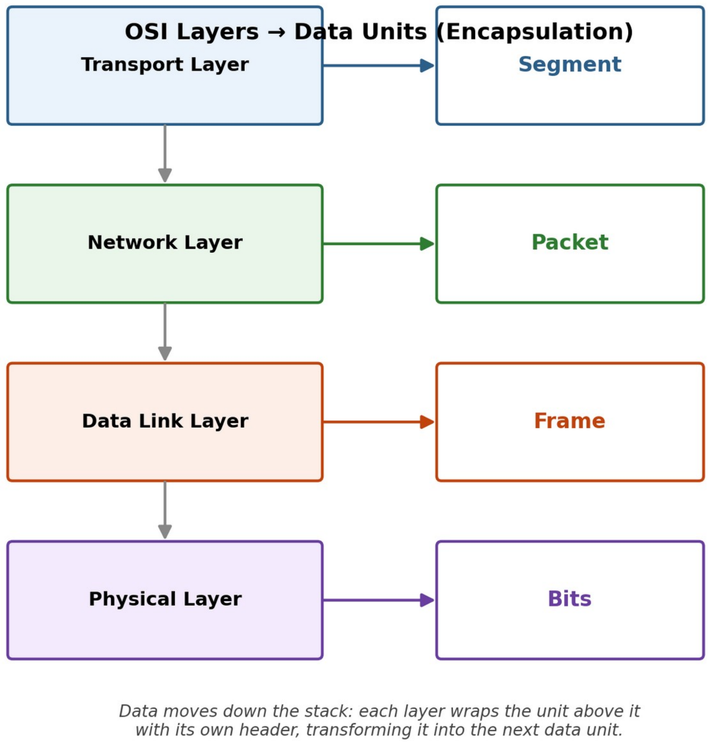
CONCEPT MAPPING

Concept Map:



1. Answer: The missing layers are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical): Application → Presentation → Session → Transport → Network → Data Link → Physical. The Session Layer establishes, manages and terminates the dialog (session) between communicating applications and provides synchronization (checkpointing), which is essential for coordinated, orderly communication. The Data Link Layer provides node-to-node framing, MAC addressing, and error detection on each physical link, which is essential before data can be converted into raw bits and sent by the Physical Layer. Without these two layers the map would break — there would be no session coordination between endpoints and no reliable hop-by-hop framing before transmission — so both are essential for reliable end-to-end communication.

Concept Map:



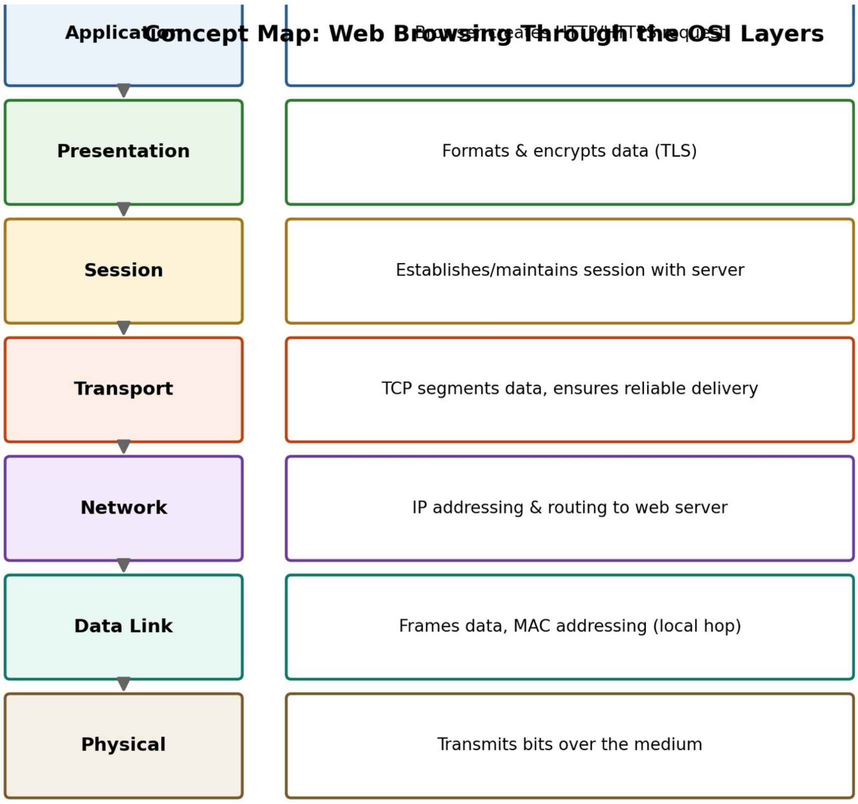
2. Answer: Concept map: Transport Layer → Segment; Network Layer → Packet; Data Link Layer → Frame; Physical Layer → Bits. As data moves down the OSI stack each layer encapsulates the unit from the layer above by adding its own header (and sometimes a trailer): the Transport Layer breaks the data stream into Segments and adds port/sequence information;

the Network Layer wraps each segment into a Packet and adds logical (IP) addressing for routing; the Data Link Layer wraps each packet into a Frame and adds physical (MAC) addressing plus error-checking for local delivery; the Physical Layer converts each frame into a stream of Bits for transmission over the medium. This step-by-step transformation (encapsulation) ensures every layer adds the control information it needs, and the receiver reverses the process (decapsulation) to recover the original data, together supporting reliable transmission.

3. Answer: The Data Link Layer uses MAC (physical) addresses to deliver frames within the same local network segment, i.e. hop-to-hop local delivery, while the Network Layer uses IP (logical) addresses to deliver packets across different networks, enabling global, end-to-end delivery through routing. Routers use IP addresses to determine the overall path a packet takes across multiple networks (global delivery), while at each individual hop along that path, MAC addresses are used to deliver the frame to the next device (local delivery). This layered addressing scheme — logical addressing for the overall journey and physical addressing for each local hop — ensures that data reaches both the correct network and the correct device within that network, achieving complete data delivery across networks.

4. Answer: The Data Link Layer performs error detection at each individual link using techniques such as CRC (Cyclic Redundancy Check) or checksums, which catch bit errors introduced during transmission over that link. The Transport Layer performs error correction on an end-to-end basis using acknowledgments, sequence numbers, and retransmission of lost or corrupted segments (as in TCP). The interaction between the two improves reliability because even if a corrupted frame is detected and discarded at one hop by the Data Link Layer, the Transport Layer's end-to-end tracking will notice the missing segment and request retransmission, so the two layers together provide a two-tier reliability mechanism: local, per-hop error detection plus global, end-to-end error correction.

Concept Map:



A break at any single layer (e.g. no Physical link, or a Transport connection failure) stops the whole chain — the page will not load.

5. Answer: Concept map for web browsing: Application Layer (browser generates an HTTP/HTTPS request) → Presentation Layer (formats and encrypts data, e.g. TLS) → Session Layer (establishes and maintains the session between browser and web server) → Transport Layer (TCP segments the data and provides reliable delivery) → Network Layer (IP addressing and routing to the web server) → Data Link Layer (framing and MAC addressing for local delivery) → Physical Layer (transmission of bits over the medium). Because each layer depends on the services of the layer below it, a failure in any single layer breaks the entire communication process: for example, if the Physical Layer link fails no bits are transmitted so nothing above it can function, and if the Transport Layer connection fails the reliable delivery needed by the Session and Application Layers is lost even though the lower layers are working — in either case the web page will not load.

6. Answer: Since the Physical and Data Link layers are working (the physical link and local frame delivery are fine) but the Network Layer fails, the problem most likely lies in IP addressing, routing, or a misconfigured gateway/router — packets cannot be routed beyond the local network. This explains why the Transport and Application layers also show failures: they depend on the Network Layer to deliver packets, so the fault cascades upward through the stack. The layered concept map helps troubleshooting by letting an engineer test bottom-up and isolate the fault to the exact layer where the chain breaks, rather than guessing across the whole stack; here it correctly directs attention to Network Layer issues (such as IP configuration, routing tables, or the default gateway) instead of wasting time checking cables (Physical Layer) or the application software, since those layers are already confirmed to be working.